The Greedy Algorithm

Let $\mathcal{I} \subseteq 2^E$ be down-monotone, i.e.,

$I \in \mathcal{I},\ J \subseteq I \Rightarrow J \in \mathcal{I}$.

<u>Problem</u>: $\max: f(I)$ for f <u>submodular</u>.

$\text{st}: I \in \mathcal{I}$

<u>Greedy</u>: $S \longleftarrow \phi,\ A \longleftarrow \phi$

<u>Iterate</u>: $A \longleftarrow \{e : S \cup e \in \mathcal{I}\}$

If $A \neq \phi$

$$e \longleftarrow \underset{e' \in A}{\text{argmax}} :\ f_S(e') = f(S+e') - f(S)$$

$$S \longleftarrow S \cup e$$

<u>Exercise</u>: If $\mathcal{I}$ is a matroid and $f(S) = \sum_{i \in S} w_i$ //

then this is the greedy alg. and as we saw, it is optimal.

Performance of Greedy

In general: Greedy is not optimal. But, we have:

Thm: If $\mathcal{I}$ is the uniform matroid: $I \in \mathcal{I} \Leftrightarrow |I| \leq k$
and if f is submodular and monotone, then Greedy is $(1 - 1/e)$ - approximation.

i.e., if $OPT = \max: f(S)$
st: $S \in \mathcal{I}$

and if greedy outputs $\hat{S}$, then

$$f(\hat{S}) \geq (1 - 1/e) \cdot OPT.$$

Proof — 1/2

Let S_i = first i elts selected by Greedy.

Hence: Greedy outputs $\hat{S} = S_k$.

Let C = opt sol'n : $f(C) = OPT$.

Proof by induction on $0 \le i \le k$:

$$f(C) - f(S_i) \le (1 - 1/k)^i \cdot \overset{\mathrm{OPT}}{\overset{\downarrow}{f(C)}}$$

At step i, Greedy selects e_i that maximizes : $f_{S_{i-1}}(e)$

Claim: $f(C) - f(S_{i-1}) \le \sum_{e \in C \setminus S_{i-1}} f_{S_{i-1}}(e)$,

Indeed: $C \setminus S_{i-1} = \{e_1, \dots, e_\ell\}$, $\ell \le k$

$$f(C) - f(S_{i-1}) \underset{\text{monotone}}{\le} f(C \cup S_{i-1}) - f(S_{i-1}) = \sum_{m=1}^{\ell} \underbrace{f\Big(S_{i-1} \cup \bigcup_{j=1}^{m} e_j\Big) - f\Big(S_{i-1} \cup \bigcup_{j=1}^{m-1} e_j\Big)} \underset{\text{submodular}}{\le} \sum_{m=1}^{\ell} f_{S_{i-1}}(e_m)$$

Proof — 2/2

$$(\star)\quad f(C) - f(S_{i-1}) \le \sum_{e \in C \setminus S_{i-1}} f_{S_{i-1}}(e)$$

$$\Rightarrow f_{S_{i-1}}(e_i) \ge \frac{1}{|C \setminus S_{i-1}|} \cdot \sum f_{S_{i-1}}(e)$$

$$\ge \frac{1}{k}\big(f(C) - f(S_{i-1})\big)$$

$$\Rightarrow f(C) - f(S_i) = f(C) - f(S_{i-1}) - \underline{f_{S_{i-1}}(e_i)}$$

$$\le f(C) - f(S_{i-1}) - \frac{1}{k}\big(f(C) - f(S_{i-1})\big)$$

$$= (1 - 1/k)\,\underline{\big(f(C) - f(S_{i-1})\big)}$$

$$\le (1 - 1/k)^i f(C).$$

For $i = k$: $\quad f(C) - f(S_k) \le (1 - 1/k)^k f(C) \le e^{-1} \cdot f(C)$

$$\Rightarrow f(S_k) \ge (1 - 1/e) \cdot f(C).$$

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